

Project Proposal Group 9 for DS340W

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September 15, 2025

Abstract—This project proposes the development of a real-time sports analytics dashboard that displays continuously updated winning probabilities during live games. Unlike existing dashboards, our system introduces a novel interactive “what-if” simulation feature, allowing users to explore how hypothetical events such as substitutions, fouls, or scoring plays would affect the game’s outcome probabilities.

Index Terms—Sports analytics, real-time prediction, dashboard visualization, win probability, interactive simulation.

I. INTRODUCTION

THE purpose of this project is to create a data science solution that enhances the way live sports are experienced by both fans and analysts. Current trackers provide raw statistics and static win probability curves, but lack interactivity and predictive depth. Our dashboard will fill this gap by offering continuously updated winning probabilities, presented through intuitive visualizations.

Beneficiaries of this system include:

- Fans, who will gain engaging, real-time insight into how likely their team is to win.
- Sports media and commentators, who can enhance live broadcasts with predictive analytics.
- Teams and analysts, who can benchmark in-game decisions against probability forecasts.

The novel aspect of our work is the interactive “what-if” simulator, which enables users to adjust game factors and immediately see projected changes in win probability. This adds a participatory layer that does not exist in existing win probability dashboards.

In addition, we plan to leverage large language models (LLMs) to provide natural-language explanations of probability shifts and game scenarios, making analytics more accessible to fans and analysts.

A. Anticipated Technical Challenges

Several technical challenges are expected:

- 1) Real-time data ingestion from sports APIs without lag.
- 2) Synchronizing and cleaning play-by-play data streams.
- 3) Training predictive models that update probabilities in seconds.
- 4) Scaling the system to handle multiple games simultaneously.
- 5) Integrating the “what-if” module without disrupting live updates.

B. Data Science Framework

Our proposed solution will follow the data science pipeline:

- 1) Data acquisition - Collecting live data via sports APIs
- 2) Data preprocessing - Cleaning and aligning game events by time
- 3) Modeling - Training machine learning models (regressions, gradient boosting, neural networks, etc) on historical data in order to predict various outcome probabilities
- 4) LLM - Alongside predictive modeling, we will incorporate LLMs trained/fine-tuned on sports commentary data to automatically generate contextual insights. These models will translate raw probabilities and what-if scenarios into natural-language summaries, enabling users to understand not just the numbers but the reasoning behind them.
- 5) Integration of our novel aspect - Create and embed the what-if simulator for user interaction
- 6) Visualization - Development of the dashboard to show live odds and scenario outcomes
- 7) Deployment - Host on an advanced cluster in order to handle the stream of live data

II. IMPLEMENTATION PLAN

We will leverage Python libraries such as pandas for data pre-processing, and scikit-learn to develop baseline models. Data storage and synchronization will be managed through structured databases to ensure scalability. Evaluation of early models will be benchmarked against naive baselines, such as assuming the team currently leading will win, to demonstrate clear improvement.

A. Week 6 Milestone

Data Infrastructure

- 1) Establish a functional data pipeline ingesting live and data via NBA, MLB, NHL, and NFL APIs
- 2) Implement real time data collection from multiple concurrent games with error handling
- 3) Create a standardized schema for player statistics, team metrics, and game events
- 4) Conduct exploratory data analysis on 2022-2025 historical data across the four leagues
- 5)

Initial Models

- 1) Implement baseline logistic regression for win probability with a target of 60% accuracy

- 2) Develop a linear regression model for score prediction
- 3) Create model evaluation framework and performance tracking

Novel Aspect Integration

- 1) Create "what-if" simulator prototype allowing for parameter adjustments

B. Week 9 Milestone

Advanced Modeling

- 1) Deploy ensemble models combining regression, gradient boosting, and neural networks
- 2) Integrate advanced features such as player fatigue, home advantage, referee adjustments, etc
- 3) Deploy LLM-based module to generate automated game summaries, commentary, and explanations of 'what-if' scenario outcomes.
- 4) Enable chatbot-style interaction within the dashboard, allowing users to ask natural-language questions

Enhanced Novel Aspect Integration

- 1) Embed "what-if" simulator into main system architecture
- 2) Enable transition between live monitoring and "what-if" scenario exploration

Visualization Development

- 1) Complete dashboard displaying real-time probability updates.

System Performance

- 1) Demonstrate processing of 100+ live data points per minute across multiple games
- 2) Validate real time prediction updates and dashboard responsiveness

III. CONCLUSION

By the end of the project, we will deliver a scalable, real-time sports analytics dashboard with both predictive and interactive components. This system not only enhances fan engagement but also contributes a novel feature to sports data science: a dynamic simulator that merges live analytics with hypothetical exploration. Looking ahead, we envision our system not only serving as a fan engagement tool but also as a foundation for advanced decision support systems in sports analytics. By bridging live predictive modeling with interactive simulation, this project sets the stage for future innovations in real-time strategy optimization, broadcast enhancements, and even sports betting integrity analysis. The integration of LLMs ensures that our dashboard communicates results in a human-readable way, bridging the gap between complex predictive analytics and fan engagement. This capability positions the project as not only a quantitative tool but also a narrative platform for real-time sports analysis.

REFERENCES

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Tim Damasco - Contributed to the writing of the abstract, introduction, technical challenges, data science framework, and conclusion sections.

Brandon Barber - Contributed to the writing of the implementation plan.

Mohammad Islam - Contributed to the development of the project idea/initial research and the writing of the data science framework